One-summability of the two divergent series in every non-singular direction

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned journal source file sources/sigma-2025-020-non-integrability-of-the-sasano-system-of-ty/pinned-original-source.tex (Non-Integrability of the Sasano System of Type D5(1) and Stokes Phenomena); no statement or proof step was rewritten, reordered or removed.

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Editorial note. Journal-unit assembly: card-local preamble (author theorem environments and macros verbatim), article class, bibliography backend none; any literature citation carried by the delivered span is disclosed in the notes of this card.

Editorial note. Journal numbering seed recomputed from the source itself; printed number 1.18 per census v3.7 is only a cross-check (section 8.8.98).

Editorial note. Revision r002: regenerated from the same spec after the r001 assembly crashed during packaging.

Editorial scope. the counted statement and its own proof environment only

Allowed standard prerequisites (editorial).

- The divergent power series phi-hat and phi-hat introduced by Propos.... Supplies background that the delivered proof uses. The entry is declared as a dependency; it is not re-proved on the card, and the load-bearing steps of the counted statement are carried out inside the delivered span.

Source reference. Original reference 3.1: sources/sigma-2025-020-non-integrability-of-the-sasano-system-of-ty/pinned-original-source.tex, source lines 651–651. Navigation pointer invoked at line(s) 858: the counted text refers to this environment of the same journal unit, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the statement inside that section, derived from the shared section-scoped theorem counter of the source); the cited item is an allowed input and none of it is reproduced inside the span.

Lemma 1.18. *Under the above notations, we have*

1. *Assume that $\alpha \notin \mathbb{N}$. Then for every direction $\theta \neq \pi$ the function*

$$\varphi_\theta(\tau) = 2(\alpha-1)\int_0^{+\infty \mathrm{e}^{\mathrm{i}\theta}} (1+\zeta)^{\alpha-2}\mathrm{e}^{-\frac{\zeta}{\tau}}\mathrm{d}\zeta$$

   defines the 1-sum of the power series $\hat{\varphi}(\tau)$ in such a direction.

2. *Assume that $\alpha \notin \mathbb{Z}_{\leq 0}$. Then for every direction $\theta \neq 0$ the function*

$$\phi_\theta(\tau) = \int_0^{+\infty \mathrm{e}^{\mathrm{i}\theta}} (1-\zeta)^{-\alpha}\mathrm{e}^{-\frac{\zeta}{\tau}}\mathrm{d}\zeta$$

   defines the 1-sum of the power series $\hat{\phi}(\tau)$ in such a direction.

When $\mathrm{Re}(\alpha) \leq 2$ (resp. $\mathrm{Re}(\alpha) \geq 0$) the function $\varphi_\theta(\tau)$ (resp. $\phi_\theta(\theta)$) is a holomorphic function in the open unlimited disc

$$\mathcal{D}_\theta = \left\{\tau \in \mathbb{C} \mid \mathrm{Re}\left(\frac{\mathrm{e}^{\mathrm{i}\theta}}{\tau}\right) > 0\right\}$$

for any direction $\theta \neq \pi$ (resp. $\theta \neq 0$). Otherwise, they are holomorphic functions in the open bounded disc

$$\mathcal{D}_\theta(1) = \left\{\tau \in \mathbb{C} \mid \mathrm{Re}\left(\frac{\mathrm{e}^{\mathrm{i}\theta}}{\tau}\right) > 1\right\}.$$

Proof. Consider the divergent power series $\hat{\varphi}(\tau)$ and $\hat{\phi}(\tau)$ defined by Proposition 3.1. For the corresponding formal Borel transforms, we obtain the power series

$$\hat{\mathcal{B}}_1\hat{\varphi}(\zeta) = 2\sum_{n=0}^{\infty}(\alpha-1)_{n+1}\frac{\zeta^n}{n!} = 2(\alpha-1)\sum_{n=0}^{\infty}(\alpha-2)_n\frac{\zeta^n}{n!}, \qquad \hat{\mathcal{B}}_1\hat{\phi}(\zeta) = \sum_{n=0}^{\infty}\alpha^{(n)}\frac{\zeta^n}{n!}.$$

Both of the series $\hat{\mathcal{B}}_1\hat{\varphi}(\zeta)$ and $\hat{\mathcal{B}}_1\hat{\phi}(\zeta)$ are convergent near the origin of the Borel ζ-plane with finite radiuses of convergence. Therefore, both of the divergent series $\hat{\varphi}(\tau)$ and $\hat{\phi}(\tau)$ are Gevrey series of order 1. The functions

$$\varphi(\zeta) = 2(\alpha-1)(1+\zeta)^{\alpha-2}, \qquad \phi(\zeta) = (1-\zeta)^{-\alpha}$$

present the sums of the power series $\hat{\mathcal{B}}_1\hat{\varphi}(\zeta)$ and $\hat{\mathcal{B}}_1\hat{\phi}(\zeta)$, respectively.

Consider the function $(1+\zeta)^{\alpha-2}$. We have that $\left|(1+\zeta)^{\alpha-2}\right| = A|1+\zeta|^{\mathrm{Re}(\alpha)-2}$, where $A = \mathrm{e}^{-\mathrm{Im}(\alpha)\arg(1+\zeta)}$. Let $\theta = \arg(\zeta)$. If $\mathrm{Re}(\alpha)-2 \leq 0$, then

$$\frac{A}{|1+\zeta|^{2-\mathrm{Re}(\alpha)}} \leq A$$

when $\cos\theta \geq 0$, while

$$\frac{A}{|1+\zeta|^{2-\mathrm{Re}(\alpha)}} \leq \frac{A}{|\sin\theta|^{2-\mathrm{Re}(\alpha)}}$$

when $\cos\alpha < 0$. If $\mathrm{Re}(\alpha) - 2 > 0$, then

$$A|1+\zeta|^{\mathrm{Re}(\alpha)-2} \leq B\mathrm{e}^{|\zeta|}$$

for an appropriate constant $B > 0$. Therefore, the function $\varphi(\zeta)$ is of exponential size at most 1 at ∞ along any direction $\theta \neq \pi$ from 0 to $+\infty\mathrm{e}^{\mathrm{i}\theta}$. Moreover, the function $\varphi(\zeta)$ is continued analytically along any such a direction. Then the corresponding Laplace transform $(\mathcal{L}_\theta\varphi)(\tau)$ is well defined and gives the 1-sum of the divergent power series $\hat{\varphi}(\tau)$ in such a direction. If we denote by $\varphi_\theta(\tau)$ this 1-sum, we have that

$$\varphi_\theta(\tau) = 2(\alpha-1)\int_0^{+\infty\mathrm{e}^{\mathrm{i}\theta}} (1+\zeta)^{\alpha-2}\mathrm{e}^{-\frac{\zeta}{\tau}}\mathrm{d}\zeta$$

for every $\theta \neq \pi$. From the above estimates, it follows that when $\mathrm{Re}(\alpha) - 2 \leq 0$, the 1-sum $\varphi_\theta(\tau)$ is a holomorphic function in the open unlimited disc

$$\mathcal{D}_\theta = \left\{\tau \in \mathbb{C} \mid \mathrm{Re}\left(\frac{\mathrm{e}^{\mathrm{i}\theta}}{\tau}\right) > 0\right\}, \tag{1.1}$$

whose opening is π. When $\mathrm{Re}(\alpha) - 2 > 0$, the 1-sum is a holomorphic function in the open bounded disc

$$\mathcal{D}_\theta(1) = \left\{\tau \in \mathbb{C} \mid \mathrm{Re}\left(\frac{\mathrm{e}^{\mathrm{i}\theta}}{\tau}\right) > 1\right\} \tag{1.2}$$

with opening $< \pi$.

Using similar arguments, we find that for any direction $\theta \neq 0$ the Laplace transform

$$\phi_\theta(\tau) = \int_0^{+\infty\mathrm{e}^{\mathrm{i}\theta}} (1-\zeta)^{-\alpha}\mathrm{e}^{-\frac{\zeta}{\tau}}\mathrm{d}\zeta$$

defines the 1-sum of the divergent power series $\hat{\phi}(\tau)$ in such a direction. When $\mathrm{Re}(\alpha) \geq 0$, the 1-sum $\phi_\theta(\tau)$ is a holomorphic function in the disc $\mathcal{D}_\theta$ from (1.1). Otherwise, the 1-sum is a holomorphic function in the disc $\mathcal{D}_\theta(1)$ from (1.2). □